

DEPARTMENT: EVENTS AND SIGHTINGS

Source Code Exhibition and Software Heritage's 10th Anniversary

On 28 January 2026, the Source Code Exhibition project was unveiled at UNESCO headquarters in Paris, France, for the 10th anniversary of the Software Heritage initiative, drawing hundreds of participants from across the globe.

Software Heritage is an initiative launched at Inria (the French National Institute for Research in Digital Science and Technology) 10 years ago to build a universal archive of source code. From its early stages, it has been supported by UNESCO, which views "software source code as part of the heritage of our knowledge societies" [1]. As of today, tens of billions of unique source files have already been archived.

Institutions around the world face the challenge of preserving our computing heritage, with particular difficulty when it comes to preserving and presenting software. Some institutions, such as the Computer History Museum in Ljubljana, Slovenia, focus on rerunning historical software on original machines, while other initiatives aim to collect source code, documentation, and oral histories. This approach is exemplified by the Software Preservation Group hosted by the Computer History Museum in Mountain View, CA, USA [2].

Drawing on the Critical Code Studies movement inaugurated by digital literature scholar Mark Marino [3], for whom "source code is a text, and reading it as a cultural object means considering it in all its history, uses, interconnections, and dependencies with other code, hardware, and cultures," this exhibition aimed to treat source code as the main attraction rather than merely as background material.

The organizers¹ launched a call for contributions inviting participants to submit any piece of source code they considered interesting, accompanied by a short commentary. The submitted code could be their own or any publicly available piece of code that held relevance for them. The call was deliberately open: Contributors could be historians, programmers, artists, activists, or any combination. The exhibition committee received submissions spanning programming languages, decades, and continents. They ultimately selected 15 pieces organized around three interpretive axes: source code as historical testimony, mirror of society, and cultural artifact.

Each exhibition panel featured the source code itself alongside commentary reviewed by the exhibition committee, as well as a short biography of the contributor highlighting their relationship to the code (Figure 1). The panel design reflects the content of the code, but deliberately avoids a technical, code-screen aesthetic, favoring instead a graphic approach more suited to a museum setting.

SOURCE CODE AS HISTORICAL TESTIMONY

The first section brings together five panels spanning just over 25 years of computing history, from Turing's Manchester Mark II in 1950 to the Latin American World Model of 1976. Each invites the reader to ask not only what a program does, but what it reveals about the world in which it was written.

The exhibition includes an extract from ELIZA, written by Joseph Weizenbaum at the Massachusetts Institute of Technology (MIT) in the mid-1960s. As Team

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¹The exhibition was organized by the present authors together with Stefano Penge (Codexpo.org) and Camille Picard (Science Outreach Manager, Inria), with an International Exhibition Committee comprising David C. Brock (Computer History Museum, Silicon Valley), Pierre Depaz (Universität Basel), Stefano Penge, Valérie Schafer (Luxembourg Centre for Contemporary and Digital History, University of Luxembourg), Boštjan Špetič (Computer History Museum Slovenia), and Artemis Yagou (Deutsches Museum, Munich). The panel design was made by Vincent Devillard (vincentdevillard.com).



FIGURE 1. Guided tour of the Source Code Exhibition, UNESCO, Paris, 28 January 2026. Courtesy of the Computer History Museum Slovenia.

ELIZA [4] documents, the code recovered from the MIT archives in 2021 contains architectural choices unusual for their period, among them a CHANGE function that allows users to edit scripts while the program was running, gesturing toward what would become a defining concern of conversational artificial intelligence (AI): the user-modifiable system.

As Len Shustek argued, “source code provides a view into the mind of the designer:” Through it, “we learn what use was anticipated, which unusual cases are handled, where efficiency was thought to matter” [2]. The remaining panels demonstrate this across different registers. Kate Mancey’s panel [5] on Turing’s debugging notes for the Manchester Mark II shows how early programmers reached for musical vocabulary (intervals, pitches, harmonics) to render a computational process perceptible, revealing how they conceptualized the dynamic status of the machine they worked with. The Fortran code of the Latin American World Model described by Fenwick McKelvey and Markus E. Ramsauer [6] shows that programs are never politically neutral: Here, optimization algorithms were deployed to argue for alternative development pathways rather than inevitable resource constraints.

These three panels, alongside two further examples in the section, make the case that source code constitutes historical evidence on its own.

SOURCE CODE AS MIRROR OF SOCIETY

The act of programming is a socially embedded practice that constantly interacts with social structures,

hierarchies, and power struggles, as is well exemplified in historian Nathan Ensmenger’s work on the emergence of the programmer profession and its struggle for recognition [7]. Source code reflects how programmers sometimes adapt to corporate cultures and constraints or, conversely, resist regulations and laws perceived as arbitrary. Code also transforms the way we interact with one another, as shown in large-scale open source software projects, and with our environment.

Stéphane Bortzmeyer’s panel [8], for instance, presents a few lines of Perl that were classified as munitions by the U.S. government. The code was printed on a T-shirt as a deliberate political provocation: If cryptography is a weapon, what happens when someone wears it through airport security? This example makes the “mirror of society” argument in its sharpest form—the code’s meaning is inseparable from its legal and political context.

While nearly all exhibition panels display actual source code, one of them shows a different item: a prompt serving as input for a large language model. Large language models are massively disrupting the workplace and shifting power balances across many professions, for better and for worse. Sofia Greggio’s [9] contribution presents an alternative vision of generative AI, in which the system is instructed to ask open, reflective questions rather than providing direct answers, thus engaging the user in an interaction with ecological data about the Venice lagoon.

SOURCE CODE AS CULTURAL ARTIFACT

Computing and programming have historically been employed as means to increase human productivity and capabilities—particularly in military contexts, from the use of computers for ballistic calculation and cryptography during World War II to Defense Advanced Research Project Agency-funded developments in AI and intercomputer communication decades later. Yet code can also become a place of self-expression and creativity [10], escaping or *détourning* the purposes for which it was conceived. Code can be a cultural form, exploiting programming constraints as a source of creativity, as illustrated by the Obfuscated C Contest [11], which challenges participants to write the most obfuscated yet functional program. Code can also become a means of political expression, whether as a tool for feminist struggles or in service of a decolonial agenda.

One exhibition panel features Wenyan [12], a programming language whose syntax borrows from

Wényán—the classical written form of Chinese—to implement an algorithm described by the Chinese mathematician Yang Hui in the 13th century, though named Pascal's Triangle after a 17th-century French mathematician. This example highlights that code is embedded in culture and reflects the norms and biases of the context in which it was created. Programming remains dominated by predominantly White Western males [13], where non-English programming languages remain curiosities.

CONCLUSION

The Source Code Exhibition did not end at UNESCO. Following its January 2026 premiere, the exhibition has traveled—and continues to travel—to venues across Europe, reflecting the broad interest it has generated across academic, cultural, and technical communities. A dedicated website accompanies the traveling exhibition, making all 15 panels freely accessible online to anyone unable to attend in person [14]. What began as a project to mark the 10th anniversary of Software Heritage has since grown into a wider conversation about how we read, preserve, and make sense of the code that shapes our world.

The traveling exhibition has already reached its first venues. From 7 to 25 March 2026, it was displayed at the Carrefour Numérique² of the Cité des Sciences et de l'Industrie, Paris, where its opening coincided with the celebrations of Wikipedia's 25th anniversary during the Premier Samedi du Libre XXL—a public event bringing together contributors from across the free software community. The exhibition then moved to the Mathematics and Informatics Research Library of Sorbonne University, Paris, where it was on display from 30 March through the end of May 2026, welcoming students, researchers, and the general public. It was then presented at the Inria Grenoble campus in France and at the Congrès de la Société Informatique de France in Nancy from 3 to 5 June 2026. Further confirmed venues include the South Tyrol Free Software Conference in Bolzano, Italy, (13–14 November 2026), Télécom Paris, University of Luxembourg, and the Campus Condorcet Humathèque in Aubervilliers, France, with additional stops at the Computer History Museum in Ljubljana, Slovenia, and the Universities of Pisa (Italy), Basel (Switzerland), and Virginia (USA) to be confirmed. With each new location, the exhibition maps the diversity of its audiences, from open source communities and scientific researchers to artists and the public, enacting its core claim: that code holds different meanings for different people.

In keeping with the values of openness that run through much of the source code on display, all exhibition content—panels, commentaries, biographical texts, and website materials—is released under a Creative Commons Attribution license (CC-BY). These materials may be freely reproduced, translated, and adapted, provided the contributors are credited. We warmly welcome any institution, educator, or community wishing to host, extend, or enrich the exhibition—whether by adding new panels, translating existing ones, or bringing the collection to new audiences. Enquiries can be directed to sourcecode-exhibit@inria.fr.

The 15 panels gathered for the Source Code Exhibition necessarily offer a partial and provisional portrait of what source code can be. We hope they nonetheless demonstrate, convincingly, that code is worth reading—not only for what it does, but for what it remembers, reflects, and imagines.

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The IEEE Commemorates Manchester Code

On 13 April 2026, the University of Manchester celebrated the award of an IEEE Milestone for the digital encoding method known as "Manchester code."¹ Although the Milestone scheme has honored many large and complex initiatives, here was a chance to recognize, as IEEE past president Tom Coughlin put it in his introduction to the accompanying meeting, "a small idea with a giant footprint" (Figure 1).

Manchester code can be described in a few words. The simplest way to send binary data electronically, as a plain string of 0s and 1s, is often unreliable in practice when a long run of 0s or of 1s appears. To avoid this,

Manchester code simply replaces the values "0" and "1", respectively, with "0, then 1" and "1, then 0" (or the other way around), halving the signal rate but providing many advantages of reliability and adaptability. The idea might have arisen anywhere; but it was, in fact, devised in 1948 by the electrical engineer F.C. Williams for the first



FIGURE 1. "I give you this plaque, which you guys paid for...": Tom Coughlin (right) presents the Milestone plaque to Duncan Ivison, president and vice-chancellor of the University of Manchester. Photograph CC BY SA Troy Astarte

¹See https://ethw.org/Milestones:Manchester_Code_1948-931949